

AXONA

Synopsis and Implications

David A. Smith · *Axona.net* · v0.1 · 2026-06-12

During the mechanical ages we had extended our bodies in space. Today, after more than a century of electric technology, we have extended our central nervous system itself in a global embrace, abolishing both space and time as far as our planet is concerned.

— Marshall McLuhan, *Understanding Media: The Extensions of Man*, 1964

What it is

Axona is a peer-to-peer substrate for addressing, routing, and publish/subscribe messaging that runs with no platform in the middle. Three properties define it. It is **self-authenticating**: a participant's identity is its own cryptographic keypair, verified directly by its peers.¹ It is **geo-aware but not geo-fenced**: node addresses carry a coarse geographic prefix that biases routing toward locality — an area code, not a wall — so traffic stays efficient and community-scaled while no one is partitioned out. And it is **infrastructurally diffuse**: a single bootstrap broker introduces new peers, but the mesh sustains itself peer-to-peer, converges its state through replication and anti-entropy, and runs in an ordinary browser or an aging phone.²

We can harden the security model, close failure modes, and tune the protocol, and each such change nudges the network's behavior. But we cannot *measure* that behavior. We can measure latency. The social consequence travels on the same wire and is invisible to us.

¹ No certificate authority, no registrar, no identity broker to petition, surveil, or revoke through. You are your key.

² The entire cost of entry: a self-generated key and an open browser tab — no install, no account, no server.

What is genuinely new

We have built censorship-resistant media before — onion routing, content-addressed storage, gossip social networks. Axona's pub/sub layer extends that lineage with a sharper consequence: it furnishes **ungovernable mobilization and un-censorable assembly**. A topic cannot be deplatformed, because there is no platform; a message cannot be reliably unsaid, because retraction is best-effort and any holder of the bytes keeps them; a movement cannot be switched off at a chokepoint, because the chokepoint does not exist. For people organizing under hostile infrastructure — dissidents, mutual-aid networks, civil-defense communications when centralized systems fail or are seized — this is a profound civic good. Assembly is among the

oldest of liberties, and Axona returns it to a form no intermediary can withdraw.

The qualitatively new element is *who is on the network*. When the publishers and subscribers are largely AI agents rather than people, Axona becomes the medium in which **machine-speed information cascades occur with no platform standing in the middle to dampen them**. Human cascades are bounded by human tempo — attention, fatigue, the friction of reading and deciding. Agent cascades are not. A signal, a rumor, a coordination instruction can propagate across a swarm and trigger action faster than any human, institution, or regulator can observe it, let alone intervene.³

The dual-use reality

The properties that shelter the dissident shelter the harasser; the assembly that cannot be censored is also the assembly that cannot be moderated; the cascade that cannot be dampened carries disinformation as readily as a flood warning. This is not a flaw to be engineered away — it is the *same property* seen from two sides. **Un-censorable and un-governable are one fact**. To build a medium that no power can silence is to build a medium that no power can stop, including powers we would want stopped, and including non-human actors operating at a tempo our institutions were never designed to match.

Why we proceed, and what we owe

We build it because the alternative — assembly and speech held permanently at the discretion of a few intermediaries, now augmented by AI that those same intermediaries control — is its own grave risk. A neutral, self-owned substrate is a hedge against that concentration. But neutrality is not innocence. The responsibility that platforms discharged badly does not vanish when the platform does; it devolves onto the architecture, onto the applications built atop it, and onto the norms of the communities — human and agentic — that inhabit it. Our obligation is to name this plainly, to shape the affordances that govern a cascade — the cost of an identity, the pull of proximity, the reach of a retraction, the rate of spread — with that cascade in mind, and to keep studying the process we have set loose.

*Axona is not a wiring diagram. Axona is a process. It is defined less by its topology than by what flows through it — and that is no longer on paper. **The first version is in the wild; anyone can use it.***

³ Markets already gave us the flash crash as a preview. Axona generalizes that dynamic from prices to arbitrary coordination — and removes the one thing every prior platform, however biased or captured, also happened to supply: a brake.